WEEK 1 — EDA & EXECUTIVE INSIGHTS MEMO

Project: Attenomy Telemetry Anomaly Analysis | Task 1.3 | Prepared from supplied EDA CSV outputs

1. Executive Summary

The cleaned telemetry dataset contains 25,000 records. The analysis split comprises 17,500 training records, 3,750 validation records, and 3,750 test records. In the training split, 685 records are labelled anomalous (3.914%); therefore, the target is imbalanced. Nine statistical tests were reported. At the Bonferroni-adjusted threshold of 0.005556, four tested features show statistically significant evidence of association or distributional difference with the anomaly label: **error_code**, **retry_count**, **payload_size_kb**, and **latency_ms**. These are associations, not proof of causation. The findings support proceeding to leakage-aware classical baselines in Week 2.

2. Dataset & Analysis Scope

Metric	Value
Total records	25,000
Train / validation / test	17,500 / 3,750 / 3,750
Training anomalies	685 (3.9143%)
Hypothesis tests / adjusted alpha	9 / 0.005556

3. Statistical Hypothesis Testing

For categorical features, the null hypothesis states independence from *is_anomaly*; for numeric features, the null hypothesis states equal distributions across anomaly labels. The adjusted decision threshold is $\alpha = 0.05 / 9 = 0.005556$.

Feature / test	Statistic (df)	p-value	Effect size	Decision
error_code Chi-square	1426.2115 (5)	2.878e-306	Cramer's V 0.2855	Significant
retry_count Mann–Whitney U	6,679,158	5.001e-17	Rank-biserial 0.1597	Significant
payload_size_kb Mann–Whitney U	6,746,490	2.578e-14	Rank-biserial 0.1714	Significant
latency_ms Mann–Whitney U	6,146,453.5	0.002804	Rank-biserial 0.0673	Significant
channel Chi-square	3.8692 (3)	0.275934	Cramer's V 0.0149	Not significant
interaction_type Chi-square	3.9615 (4)	0.411239	Cramer's V 0.0150	Not significant
session_duration_sec Mann–Whitney U	5,655,143.5	0.422326	Rank-biserial -0.0181	Not significant
region Chi-square	1.6727 (3)	0.643013	Cramer's V 0.0098	Not significant
client_tier Chi-square	1.7500 (4)	0.781624	Cramer's V 0.0100	Not significant

Interpretation: the four significant results merit further investigation as potential predictors. Non-significant results do not prove that no relationship exists. The Chi-square minimum expected frequencies reported in the CSV are all above 5, supporting the usual expected-count condition for these tests.

OPERATIONAL IMPLICATIONS & WEEK 2 PLAN

Continuation of Week 1 EDA Executive Memo

4. Empirical Hypotheses — Summary

H1: error_code and is_anomaly are associated (Chi-square; supported).
H2: retry_count distributions differ between anomaly and normal records (Mann–Whitney U; supported).
H3: payload_size_kb distributions differ by anomaly label (Mann–Whitney U; supported).
H4: latency_ms distributions differ by anomaly label (Mann–Whitney U; supported).
H5: channel and is_anomaly are associated (Chi-square; not supported at adjusted α).
The notebook also tested interaction_type, session_duration_sec, region, and client_tier. Decisions use the Bonferroni-adjusted threshold; they should not be interpreted as causal findings.

5. Operational Risks & Controls

Class imbalance. Only 3.914% of training records are anomalies. Accuracy alone can be misleading. Report precision, recall, F1, confusion matrix, and precision–recall AUC; consider class weighting and threshold tuning using validation data.

Potential label leakage. error_code has a strong association with the target (Cramer's V $\approx$ 0.2855). Confirm how is_anomaly was generated and whether error_code is available at prediction time. If the target was derived from error_code, exclude it or document the task as rule replication rather than prospective detection. Make this decision before modeling.

Collinearity and redundant signals. Review the predictor correlation matrix and investigate highly correlated pairs. Fit all imputation, scaling, encoding, feature selection, and other preprocessing only on training data, then apply the fitted transformations to validation/test data. Do not use validation/test results to choose preprocessing parameters.

Operational validity. Confirm timestamps are parsed consistently, inspect whether temporal patterns can cause leakage, and keep a clear record of split strategy and random seed. If deployment predicts future events, consider a chronological split rather than relying only on random splitting.

6. Week 2 Classical Modeling Plan

1. Define the prediction point and label provenance. Document when the prediction is made and which fields are available then. Resolve the error_code leakage question before fitting models.
2. Establish baselines. Train a majority-class baseline and a simple DummyClassifier; then Logistic Regression and a tree-based baseline such as Decision Tree or Random Forest. Use a Pipeline to keep preprocessing within training folds/splits.
3. Handle imbalance responsibly. Start with class_weight='balanced' where supported; compare alternatives only on training/validation data. Do not resample before the split.
4. Evaluate. Report precision, recall, F1, PR-AUC, ROC-AUC as a secondary metric, confusion matrix, and the chosen operating threshold. Compare against the baseline and document false-positive/false-negative trade-offs.
5. Finalize. Tune using training/validation only; evaluate the selected model once on the held-out test set; save the pipeline, metrics, and a short model card.

7. Recommendations

Proceed to Week 2 only after confirming label provenance and feature availability. Prioritize leakage-safe comparisons using retry_count, payload_size_kb, and latency_ms, and assess error_code both with and without it if its use is legitimate. Treat all Week 1 statistical results as exploratory evidence; validate predictive utility through held-out evaluation. The supplied CSVs support the numerical results in this memo, but they do not by themselves establish causality or confirm every plot's visual interpretation.

Decision: Week 1 statistical outputs are available; executive memo documents findings and proposes a leakage-aware Week 2 baseline plan.